

RESEARCH TOPIC/TITLE SteelTrack: A Centralized Operations Management System for Multi-Branch Stainless Steel Enterprises using Machine Learning for Inventory Forecasting

A. Description of the Project

1. Rationale

Stainless steel manufacturers and suppliers play a key role in construction, infrastructure development, and the production of metal products like railings, gates, chairs, and other structural elements. These companies have more than one office. They buy raw materials, hold inventories, pay bills and produce goods. These processes need to be carefully synchronized so as to provide accurate inventory records, timely delivery of services and efficient use of resources. But many organizations are still on fragmented, non-integrated platforms. It impedes operations, creates wrong entries and makes it harder to track the overall performance of the business. A major hurdle in modernizing these multi-branch operations is the presence of disconnected, antiquated legacy databases that store critical historic business data in inaccessible silos. Without a technical mechanism to extract and clean this historical information, companies remain bound to obsolete systems.

A company with multiple branches needs a centralized management system for better coordination and efficiency of work. When data and operations are on a single platform, it is possible to consolidate information and track branch activities that can lead to better decision making [1]. This absence of a centralized system may result in data inconsistency, delays in updates, and difficulties in controlling operations across numerous branches [2]. In the daily routine sales processing in each branch of the stainless-steel company the updating of inventory and transfer of items is done. Poor recordkeeping, inventory problems, and inaccurate stock levels are often blamed on human or system processing [3]. Another cause of an inefficient distribution of stocks may be poor recordkeeping of transfers between branches. Research [4] emphasizes the importance of inventory management systems in supply chain operations as they improve transparency, efficiency and accuracy.

Multi-location companies also have an issue with real-time visibility. Sales, inventory, branch activity – these are the things business owners and managers need to know about on time to make sound judgments. Manual or piecemeal data collection cannot provide analysis of business performance or quick reaction to operational problems. Modern management systems have emphasized the importance of central platforms for monitoring and controlling processes in real time [5]. One big problem is to keep the data consistent and synchronous across multiple branches. Simultaneous transactions from different branches [6] could cause late updates, contradictory records and incorrect information if the systems are not synchronized properly. Traditional database architectures rely on heavy, manual table queries that choke branch bandwidth and delay updates, introducing a critical need for lightweight, push-based synchronization protocols to sustain modern multi-branch operations.

The recent development of intelligent management systems further strengthens the effectiveness of the data-driven approaches for operational decision support. [7] These tools enable companies to analyze data, optimize operations and resource allocation, and get automated monitoring and recommendations. Thus, the proposed research would use an automated Extract, Transform, Load (ETL) pipeline to safely ingest disconnected historical records, a centralized database, a decoupled machine learning service, rule-based decision support, and active real-time synchronization via web protocols to significantly improve operational efficiency and inventory management. The study will also create a Data Trust

Score to assess the consistency and reliability of data shared across branches on inventories and operations, operating as an automated gatekeeper during the initial data migration phase to prevent corrupted legacy inputs from degrading system operations.

The proposed study aims to use machine learning to predict inventory by using the Long Short-Term Memory (LSTM) model, a type of recurrent neural network that works exceptionally well when dealing with time series data such as previous sales data to increase system capabilities. Unlike standard statistical methods or linear algorithms that evaluate flat trend lines, an LSTM architecture contains specialized hidden layers and memory cells designed to retain long-term dependencies and context within sequential data. This mathematical capability enables the framework to autonomously recognize non-linear patterns, cyclical business operations, and complex multi-branch supply disruptions. The system will forecast the future demand of the product for each branch using historical transaction data such as product quantities, frequency of items sold and branch performance. It will also help to identify the seasonal trends and peak demand period, especially in the busy months of July and December and classify the products as fast-moving and slow-moving depending upon their sales pattern. Such predictive insights will be critical for operational decision-making, from procurement planning, production scheduling and inventory allocation across branches to avoid overstocking and stock outs, increase resource utilization and drive overall business performance.

The authors have presented SteelTrack: An Intelligent Centralized Operations Management System for Multi-Branch Stainless Steel Enterprises to address these problems. The technology plans to place sales, inventory and branch operations onto a single platform to allow for centralized management, real-time syncing and improved collaboration between branches. All branches would be connected to a central database to share information and ensure that data is consistent and available. To resolve the technical challenge of handling historical records, the system implements an automated C# ETL Background Worker that programmatically establishes secure connection strings to old branch databases, extracts past sales logs, executes automated data cleaning, and loads the structured data directly into a unified PostgreSQL database. Furthermore, instead of relying on sluggish, request-heavy database polling, SteelTrack incorporates an event-driven synchronization layer powered by ASP.NET Core SignalR. This ensures that any new changes to sales, inventory, and stock transfers across any branch will be immediately pushed and made visible across all branches through websocket connections. To maximize scalability, the predictive engine is decoupled into an autonomous Python FastAPI microservice that hosts the trained LSTM network, communicating smoothly with the C# backend via secure REST API payloads. A rule-based decision support tool will also be used to support the operational decisions, i.e. stock management and inter branch transfers. The Data Trust Score will be used to evaluate the quality and consistency of inventory data. With the help of these methodologies SteelTrack will improve inventory accuracy, operational efficiency, data reliability and decision-making capabilities for multi-branch stainless steel companies.

Table 1. Comparison Matrix

Features	Traditional Manual System	Basic POS System	Standalone Inventory System	Generic ERP Systems	SteelTrack (Proposed System)
Centralized Multi-Branch Management					✓
Real-Time Data Synchronization				✓	✓
Inter-Branch Inventory Transfer					✓
Inventory Monitoring	✓	✓	✓	✓	✓
Sales Transaction Processing (POS)	✓	✓		✓	✓
Unified Operations Dashboard					✓
Intelligent Decision Support					✓
Legacy Database Migration (ETL)					✓
Real-Time Data Sync (SignalR)					✓
Deep Learning Forecasting (LSTM)					✓
Data Trust Score					✓
Automated Data Trust Validation					✓

The comparison matrix presents the key features of existing systems in relation to the proposed system, SteelTrack, showing that traditional manual systems are capable of basic inventory monitoring but lack centralized management and real-time data synchronization, which are essential for multi-branch operations. Basic point of sale systems are effective in handling sales transactions; however, they do not support centralized control and are limited in managing inventory across multiple branches. Standalone inventory systems can monitor stock levels but are not integrated with sales processes and do not support coordination between branches. Generic enterprise resource planning systems provide integrated solutions and centralized management, but they are often complex and costly and may only partially support features such as real-time synchronization and inter-branch

inventory transfer and are not specifically designed for medium-scale stainless steel enterprises. These limitations are consistent with studies emphasizing the challenges of system integration, cost, and scalability in enterprise systems [2], [5], [6]. In contrast, SteelTrack provides a complete and highly innovative integrated solution by supporting automated legacy database migration pipelines, centralized multi-branch management, real-time push synchronization via SignalR, inter-branch inventory transfer, inventory monitoring, and sales processing within a single platform, along with a dedicated deep learning forecasting microservice, rule-based decision support, and a Data Trust Score to ensure reliable and data-driven operations.

To provide decision support in preventing overstocking, supporting procurement, production planning, and allocating inventory per branch using a rule-based decision support system with forecasting-based inventory recommendations. To support data-driven decision-making in the proposed system, several computational models will be utilized. These models transform operational data into measurable outputs, which aligns with the use of predictive analytics and data-driven approaches in inventory and supply chain management [3], [6], [7].

One key measure is the Inventory Forecast Accuracy Score (IFAS), which evaluates how accurate the predicted demand is compared to actual demand.

$$IFAS = \left(1 - \frac{|AD - FD|}{AD} \right) \times 100$$

Where:

IFAS = Inventory Forecast Accuracy Score

AD = Actual Demand

FD = Forecasted Demand

Another measure is the Inventory Balance Score (IBS), which determines whether a branch has sufficient stock based on required demand.

$$IBS = \left(\frac{AS}{RS} \right) \times 100$$

Where:

IBS = Inventory Balance Score

AS = Available Stock

RS = Required Stock

The system also generates a Stock Transfer Recommendation Score (STRS), which supports inter-branch inventory allocation.

$$STRS = (ES - RS) + FD$$

Where:

STRS = Stock Transfer Recommendation Score

ES = Excess Stock

RS = Required Stock

FD = Forecasted Demand

Another important measure is the Product Movement Classification Score (PMCS), which classifies products based on sales performance.

$$PMCS = \left(\frac{SQ}{TS} \right) \times 100$$

Where:

PMCS = Product Movement Classification Score

SQ = Quantity Sold

TS = Total Stock

The system also computes a Seasonal Demand Index (SDI), which identifies peak demand periods.

$$SDI = \left(\frac{MS}{AS} \right) \times 100$$

Where:

SDI = Seasonal Demand Index

MS = Monthly Sales

AS = Average Sales

Finally, the system includes a Data Trust Score (DTS), which evaluates the reliability of system data.

$$DTS = \frac{(AC + CC + CS)}{3} \times 100$$

Where:

DTS = Data Trust Score

AC = Accuracy of Data

CC = Completeness of Data

CS = Consistency of Data

These computational models enable the system to provide accurate forecasting, efficient inventory management, and reliable decision support.

2. Objectives of the Study

2.1. General

The main purpose of the study is to develop an intelligent centralized operations management system for multi-branch stainless steel enterprises. Specifically, the study aims to integrate legacy data ingestion, optimize inventory processes, improve inventory accuracy, monitor branch operations in real time via event-driven protocols, and provide reliable decision support through a decoupled deep learning forecasting engine and decentralized rule-based scores.

2.2. Specific

The researchers specifically aim to achieve the following technical objectives:

- 1.
2.
 - a.
 - b.
 - 2.2.1. To design and implement an automated Extract, Transform, Load (ETL) migration worker in C# to securely establish database connections to old branch nodes, extract past sales tables, apply data-cleaning protocols, and safely transfer the historical records into a centralized PostgreSQL database system.
 - 2.2.2. To establish real-time data synchronization across all distributed branches using ASP.NET Core SignalR, ensuring immediate push-based

visibility of POS transactions, inventory adjustments, and stock movements on a unified dashboard without lagging local workflows.

2.2.3. To develop a Machine Learning microservice using Python via FastAPI and Keras/TensorFlow powered by a Long Short-Term Memory (LSTM) network to ingest migrated historical sequences, predict monthly product demand, isolate seasonal spikes, and classify fast-moving and slow-moving products.

2.2.4. To deploy a rule-based decision support tool that ingests the Python microservice's demand forecasts to evaluate mathematical stock scores (IFAS, IBS, STRS, PMCS, SDI) and automatically generate actionable inter-branch stock transfer recommendations to mitigate overstocking and stockouts.

2.2.5. To formulate an automated Data Trust Score (DTS) feature to programmatically calculate data quality, ensuring the consistency, accuracy, and completeness of operational data moving across the network while flagging corrupted entries before they pollute the forecasting model.

3. Conceptual Framework

This structural design follows the Input-Process-Output or IPO model, where the system receives operational and inventory data as input, processes the data through centralized synchronization, inventory monitoring, machine learning forecasting, and decision-support mechanisms, and produces outputs such as inventory forecasts, stock recommendations, real-time reports, and improved operational efficiency.

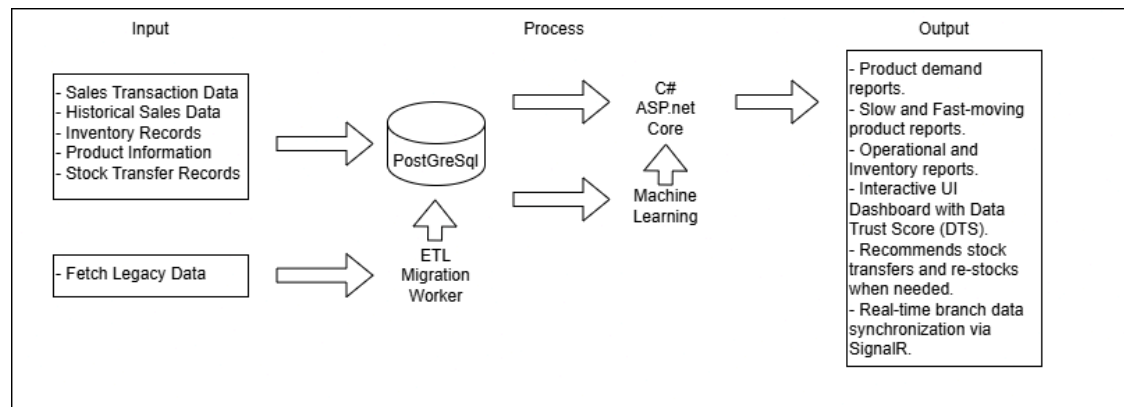


Figure 1. IPO Model

4. Methodology

The researchers will utilize the Agile Scrum Methodology in developing the proposed SteelTrack system. Agile Scrum is suitable for the study because it allows the development process to be divided into smaller phases or sprints, making it easier to monitor progress, isolate operational bugs, and continuously refine technical features. Through regular sprint reviews and validation testing, the 2-member team can ensure that the decoupled architecture behaves efficiently and addresses the core operational problems of multi-branch stainless steel enterprises. The Scrum team consists of the product owner, scrum master, and the development team. The product owner defines project parameters and refines system backlogs, the scrum master removes architectural bottlenecks, and the development team directly handles software engineering, coding, model training, and integration tasks.

The development process will consist of the following sprints:

Sprint 1: Legacy Database Fetching & Centralized Core

The development team will construct the unified, centralized PostgreSQL database schema and implement the core backend services. This sprint focuses on developing the C# ETL background worker service that safely interfaces with old, fragmented branch databases. The worker will extract legacy tables, run programmatic sanitization functions to remove incomplete variables, and load the cleaned dataset into the central database alongside basic modules for branch registration and user role permissions.

Sprint 2: Real-Time POS Integration & Push Sync

The development team will build and integrate the primary sales transactional processing and inventory adjustment components. To eliminate sluggish data fetching loops and system lag, web endpoints will be developed using ASP.NET Core Web API. This sprint introduces ASP.NET Core SignalR to create live, open socket connections, ensuring that whenever a product is sold or checked out at a branch terminal, the event instantly synchronizes and reflects on the manager's operational view.

Sprint 3: Deep Learning Prediction & Microservice Setup

The development team will construct the intelligent analytics component of the system. Rather than coupling complex mathematical modeling within C#, the predictive framework is isolated into a separate Python microservice utilizing FastAPI. This sprint involves configuring a Recurrent Neural Network using Keras/TensorFlow to deploy an LSTM prediction model. The service reads the historical timelines migrated during Sprint 1 to compute next-month item demands, spot recurring July/December business fluctuations, and categorize product velocities.

Sprint 4: Rule-Based Decision Support & Scoring Engine

The development team will establish the automated administrative logic within the C# application layer. The backend will be programmed with the core scoring formulas (IFAS, IBS, STRS, PMCS, SDI). This engine reads the real-time inventory balances from PostgreSQL and matches them against the demand outputs generated by the Python LSTM endpoint, using strict logic to automatically drop stock transfer alerts on user interfaces. Concurrently, the Data Trust Score (DTS) code is written to continuously monitor and report data reliability metrics.

Sprint 5: Full System Integration, UI Polish, and Testing (November)

The development team will tie the frontend web dashboard directly to the API endpoints, exposing the live charts, tracking grids, and forecasting modules. Rigorous software testing methodologies—including functionality validation, end-to-end load testing, and forecast accuracy auditing—will be conducted to ensure the entire multi-branch ecosystem performs reliably and securely before final defense.

Sprints:

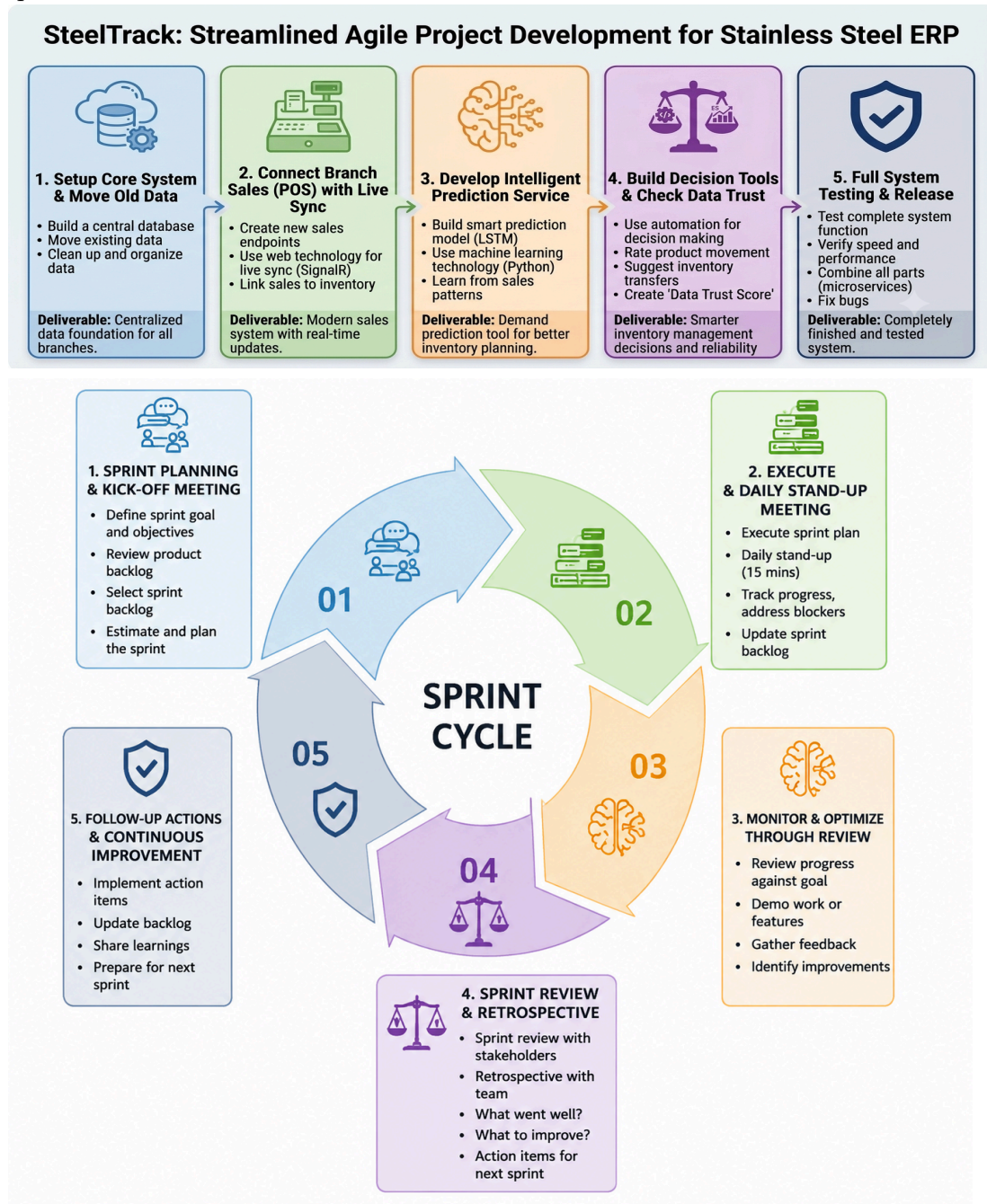


Figure 2. Agile Scrum

5. Research Project Duration

Tasks	April	May	June	July	August	Sept	Oct	Nov	Dec
Topic proposal & Approval									
Initial Research & Problem Identification									
Requirement Gathering									
Review of Related Literature									
System Analysis & Conceptual Design									
Database Design (PostGreSql)									
UI/UX Design & Wireframing									
System Integration									
System Testing & Debugging									
User Acceptance Testing									
Documentation Writing									
Final Revision & Improvement									
Final Defense Preparation									
Final Presentation & Deployment									

Figure 3. Gantt Chart

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Prepared by: Rodrigo F. Balase Jr.
Christian Jose

Date: June 11, 2026

Endorsed by:

FE B. YARA
Research Coordinator

RAMCIS N. VILCHEZ, DIT
Dean of College